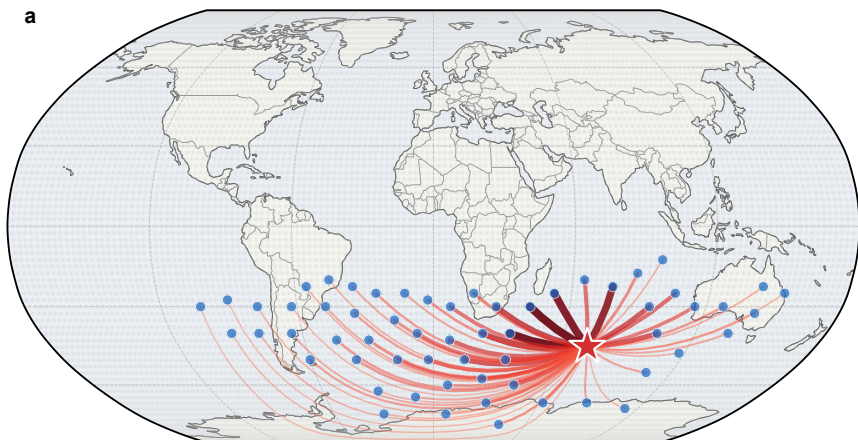


Causal Flow Pathways from Top Gateway

Source: 45°S, 72°E region (-50, 60) | N=10,512 nodes | Top-60 causal targets
(Line width ~ causal strength)

a

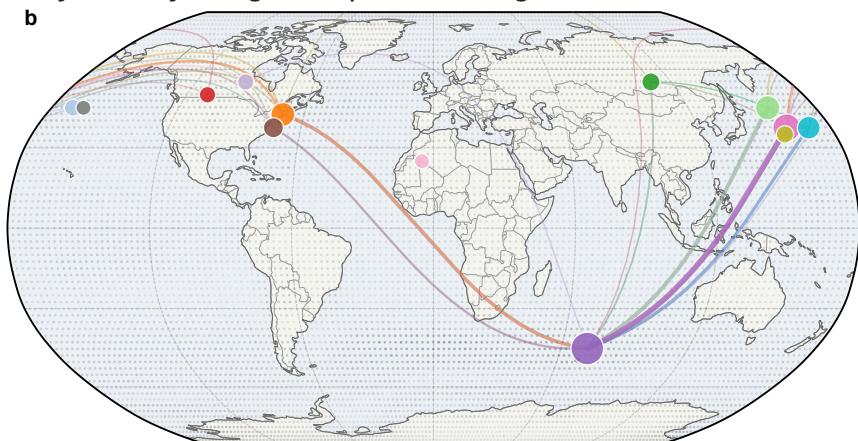


★ Source (45°S, 72°E) region (-50, 60) ● Top-60 targets — Causal pathway (width ~ strength)

Causal Hierarchical Structure

Gateway intensity background | 15 macro-region hubs & causal flow overlay

b



● Causal gateway intensity ● Region hub (size ~ strength) — Inter-region causal flow

Topological Reorganization: El Niño vs La Niña (Winter NDJF)
Difference Map — Positive/Red = Stronger Causal Gateway in El Niño

c

